

Personal details

Date of Birth 21/06/1989
Nationality German
Personal status married, one child

Education

- 02/2015– **PhD student**, *Otto-von-Guericke-University*, Magdeburg.
12/2019
 - o Title: *Prospective motion correction for high resolution gradient recalled echo-based magnetic resonance imaging at ultra-high field*
 - o Examiners:
 - Prof. Dr. rer. nat. Georg Rose
 - Prof. Dr. rer. nat. habil. Oliver Speck
 - Prof. Dr. rer. nat. Nikolaus Weiskopf
 - o Grade: summa cum laude
 - o DOI: 10.10.25673/32326
- 04/2012– **Master of Science**, *Otto-von-Guericke-University*, Magdeburg.
01/2015
 - o Medical Systems Engineering
 - o Final grade: 1.2
 - o Master thesis:
 - Title: *Analysis of a Prospective Motion Correction System for Magnetic Resonance Imaging*
 - Supervisor: Prof. Dr. rer. nat. habil. Oliver Speck
 - Grade: 1.1
- 10/2007– **Bachelor of Science**, *Otto-von-Guericke-University*, Magdeburg.
06/2012
 - o Electrical Engineering and Information Technology
 - o Final grade: 2.5
 - o Bachelor thesis:
 - Title: *Blind Source Separation for Electrocardiography during Magnetic Resonance Imaging*
 - Supervisor: Prof. Dr. rer. nat. Georg Rose
 - Grade: 1.3

Professional experience

- 03/2025– **Junior professor for *Highfield-MR Methods for Neuroimaging***, *Otto-von-Guericke-University*, Magdeburg.
Today
- 05/2022– **Visiting researcher**, *Deutsches Zentrum für Neurodegenerative Erkrankungen (DZNE)*,
Today Magdeburg.
- 02/2015– **Researcher**, *Otto-von-Guericke-University*, Magdeburg.
02/2025
 - Prospective motion correction for high resolution MRI at ultra-high field
 - Sequence development in IDEA
 - Ultra-high resolution Time-of-Flight angiography and quantitative susceptibility mapping
 - Quantitative assessment and pattern analysis of the cerebral vasculature
 - Support and supervision of several 3T and 7T MR studies
 - Third-party funding acquisition
 - Supervision and teaching
- 01/2019– **Parental leave**.
03/2019

- 01/2011– **Programmer**, G&K Medizinische Systeme, Magdeburg.
 12/2012 Development and programming of medical software in Delphi
 Support in quality management and certification of medical software
 Installation and support for a PACS and RIS

Awards & stipends

- 2022 Educational Stipend for ISMRM Workshop on Neurofluids: Anatomy, Physiology & Imaging, Rome
 2021 ESMRMB Early Career Fellowship
 2021 Young scientist at the 70th Lindau Nobel Laureate Meeting
 2020 Young scientist at the Lindau Nobel Laureate Meetings' Online Sciences Days
 2020 Merit Award, 36th Annual Scientific Meeting ESMRMB, online meeting
 2020 Magma cum laude award, 28th Annual Meeting ISMRM, virtual meeting
 2019 Participant of the 9th Nachwuchsakademie Medizintechnik (NAMT-2019) "Quantitative MRI as a key modality in life sciences"
 2019 Magma cum laude award, 27th Annual Meeting ISMRM, Montreal
 2018 Educational Stipend for Joint Annual Meeting ISMRM-ESMRMB 2018, Paris
 2017 Educational Stipend for 25th Annual Meeting ISMRM, Honolulu
 2017 Summa cum laude award, 25th Annual Meeting ISMRM, Honolulu
 2016 Educational Stipend for 24th Annual Meeting ISMRM, Singapore
 2016 Best Poster Award, 7th Annual Scientific Symposium Ultrahigh Field Magnetic Resonance, Berlin

Grants & funding

- G1 Non-invasive pressure measurement using 7T phase contrast MRI
EFRE grant from the Forschungscampus STIMULATE (under the operation number 'ZS/2023/12/182010' as part of the initiative "Sachsen-Anhalt WISSENSCHAFT Schwerpunkte")
 15.09.2025-14.02.2026 as PI
- G2 Mesoscopic imaging of human brain structure, function and neurofluids
Z02 in the CRC 1436, DFG 425899996
 01.01.2025-31.12.2028 as PI
- G3 BB-DARS: Blood biomarker DrAinage reserve score for personalized risk assessment of ARIA under Aβ immunotherapy
by the Deutsche Alzheimer Gesellschaft (DAG) e.V.
 01.10.2024-30.09.2026 as Co-PI
- G4 Quantification of perivascular spaces in neuropsychiatric long-COVID/post-COVID (LC/PC) syndrome as a biomarker for persisting perivascular inflammation and disease trajectories (JE2/TP5)
by the BMBF as part of the start phase for the German Center for Mental Health (DZPG)
 01.07.2023-30.06.2025 as Co-PI
- G5 Vascular resistance and resilience in ALS – an ultrahigh-resolution 7T MRI study of the motor cortex
DFG MA 9235/3-1; 501214112
 01.10.2022-30.09.2024 as PI
- G6 MD-DARS: MagDeburger DrAinage-Reserve-Score for patient-specific, MRI-based prediction of perivascular drainage in the Alzheimer's continuum
by the Deutsche Alzheimer Gesellschaft (DAG) e.V.
 01.10.2022-30.09.2024 as Co-PI

- G7 Vessel distance mapping: Quantification of subcortical arterial and venous vascular patterns to study their interdependency
DFG MA 9235/1-1 (NAMT9); 446268581
01.07.2020-30.06.2021 as PI

Book chapters

- B1 **Mattern H**, Lüsebrink F, Speck O.
High Resolution Structural Brain Imaging published in *Motion Correction in MR: Correction of Position, Motion, and Dynamic Field Changes* edited by van der Kouwe A, Andre J, 2022, ISBN: 9780128244609, DOI: 10.1016/B978-0-12-824460-9.00014-5

Journal publications

- J1 Arndt P, Pfister M, Müller SJ, Perosa V, **Mattern H**, Dörner M, Müller P, Meuth SG, Garz C, Neumann K, Bernal J, Schreiber S.
Perivascular spaces are associated with CSF A β in cerebral amyloid angiopathy but not in deep perforator arteriopathy
Stroke 2026
DOI: TBA
- J2 Arndt P, Khadraoui E, Müller S, Neumann K, **Mattern H**, Meuth SG, Perosa V, Charidimou A, Schreiber S.
Meningovascular inflammation in sporadic cerebral amyloid angiopathy-related cortical superficial siderosis
Annals of Clinical and Translational Neurology 2026
DOI: 10.1002/acn3.70315
- J3 Einspänner E, **Mattern H**, Großmann H, Khadhraoui E, Müller SJ, Garza AP, Dunay IR, Schregel K, Guttman CRG, Fuchs E, Behme D.
A Multimodal 7T MRI and Biomarker Study Reveals Reversible Brain Changes Following Acute Sleep Deprivation
Sleep Medicine 2025
DOI: 10.1016/j.sleep.2025.108663
- J4 Mietzner G, Lümke L, Schreiber F, Brüggemann J, Benramadan A, Al-Dubai M, Sciarra A, Knoll C, Kuehn E, Speck O, Schreiber S, **Mattern H**.
Assessing arterial patterns in the motor cortex with 7 Tesla magnetic resonance imaging and vessel distance mapping
Human Brain Mapping 2025
DOI: 10.1002/hbm.70311
- J5 Arndt P, Boewe S, Brüggemann J, Garcia-Garcia B, Yakupov R, Vockert N, Maas A, Pfister M, Perosa V, Al Dubai M, Jansen R; Meuth SG, Dörner M; Müller P, Henneicke S, Schreiber F, Neumann K, **Mattern H***, Schreiber S*.
Putamen vascularization on high-resolution 7T MRI is associated with perfusion and cognitive performance in cerebral small vessel disease
NeuroImage 2025
DOI: 10.1016/j.neuroimage.2025.121426

***equal contribution**

- J6 Arndt P, Morton L, Briesse M, Lämmlein N, **Mattern H**, Hasanuzzaman M, Westhues M, Khoshneviszadeh M, Appenzeller S, Gündel D, Toussaint M, Brust P, Kniess T, Oelschlegel A, Goldschmidt J, Meuth SG, Müller P, Braun-Dullaeus R, Debska-Vielhaber G, Vielhaber S, Becker A, Dityatev A, Henneicke S, Sendtner M, Dunay I, Schreiber S.
Vascular and neural transcriptomics reveal stage-dependent pathways to inflammation and cognitive dysfunction in a rat model of hypertension
Journal of the American Heart Association 2025
DOI: 10.1161/JAHA.125.040710
- J7 Arndt P, Pfister M, Schreiber F, Perosa V, **Mattern H**, Bernal J, Bay B, Dörner M, al-Dubai M, Swiatek V, Neyazi B, Sandalcioğlu IE, Garz C, Meuth SG, Goertler M, Vielhaber S, Neumann K, Schreiber S.
Cerebrospinal fluid A β biomarkers predict recurrent hemorrhage and identify cerebral amyloid angiopathy in patients with lobar hemorrhage
Journal of the American Heart Association 2025
DOI: 10.1161/JAHA.124.042614
- J8 Chatterjee S, Gaidzik F, Sciarra A, **Mattern H**, Janiga G, Speck O, Nürnberger A, Pathiraja S.
PULASki: Learning inter-rater variability using statistical distances to improve probabilistic segmentation
Medical Image Analysis 2025
DOI: 10.1016/j.media.2025.103623
- J9 Arndt P, Pfister M, Perosa V, **Mattern H**, Bernal J, John AC, Dörner M, Müller P, Braun-Dullaeus RC, Garz C, Nelke C, Kokott A, Jansen R, Gliem M, Meuth SG, Henneicke S, Vielhaber S, Neumann K, Schreiber S
Risk factors and clinical significance of neurodegenerative co-pathologies in symptomatic cerebral small vessel disease
Journal of Neurology 2025
DOI: 10.1007/s00415-025-13087-z
- J10 Dörner M, Pfister M, Tyndall A, von Känel R, Neumann K, Schreiber F, Arndt P, Fuchs E, Garz C, Glanz W, Butryn M, John A-C, Hildebrand A, Euler S, Hofmann AB, Machetanz L, Kirchbner J, Tacik P, Grimm A, Jansen R, Pawlitzki M, Henneicke S, Perosa P, Labeit B, Düzel E, Meuth SG, Vielhaber S, **Mattern H**, Bernal J, Schreiber S
Associations Of Inferior Frontal Sulcal Hyperintensities On Brain MRI With Cerebral Small Vessel Disease, Cognitive Function, And Depression Symptoms
Scientific Reports 2025
DOI: 10.1038/s41598-025-87493-8
- J11 Schreiber S, Arndt P, Morton L, Garza AP, Müller P, Neumann K, **Mattern H**, Dörner M, Bernal J, Vielhaber S, Meuth SG, Dunay IR, Dityatev A, Henneicke S.
Immune system activation and cognitive impairment in arterial hypertension
American Journal of Physiology - Cell Physiology 2024
DOI: 10.1152/ajpcell.00219.2024
- J12 Knoll C, Döhler J, Northall A, Schreiber S, Rotta J, **Mattern H**, Kühn E.
Age-Related Differences in Human Cortical Microstructure Depend on the Distance to the Nearest Vein
Brain Communications 2024
DOI: 10.1093/braincomms/fcae321
- J13 Xu M, Ribeiro FL, Barth M, Bernier M, Bollmann S, Chatterjee S, Cognolato F, Gulban OF, Itkyl V, Liu S, **Mattern H**, Polimeni JR, Shaw TB, Speck O, Bollmann S.
VesselBoost: A Python Toolbox for Small Blood Vessel Segmentation in Human Magnetic Resonance Angiography Data
Aperture Neuro 2024
DOI: 10.52294/001c.123217

- J14 Dörner M, Tyndall A, Hainc N, von Känel R, Schreiber F, Arndt P, Fuchs E, Garz C, Glanz W, Butryn M, Neumann K, Bernal J, **Mattern H**, John AC, Hildebrand A, Kirchebner J, Tacik P, Grimm A, Perosa V, Jansen R, Pawlitzki M, Düzel E, Meuth SG, Vielhaber S, Schreiber S.
Neuropsychiatric Symptoms And Lifelong Mental Activities in Cerebral Amyloid Angiopathy - A Cross-Sectional Study
Alzheimer's Research & Therapy 2024
DOI: 10.1186/s13195-024-01519-3
- J15 Oltmer J, **Mattern H**, Beck J, Yakupov R, Greenberg S, Zwanenburg J, Arts T, Düzel E, van Veluw S, Schreiber S, Perosa V.
Enlarged Perivascular Spaces in the Basal Ganglia are Associated with Arteries not Veins
Journal of Cerebral Blood Flow and Metabolism 2024
DOI: 10.1177/0271678X241260629
- J16 Dörner M, Seebach K, Heneka MT, Menze I, von Känel R, Euler S, Schreiber F, Arndt P, Neumann K, Hildebrand A, John AC, Tyndall A, Kirchebner J, Tacik P, Jansen R, Grimm A, Henneicke S, Perosa V, Meuth SG, Peters O, Hellmann-Regen J, Preis L, Priller J, Spruth EJ, Schneider A, Fliessbach K, Wiltfang J, Jessen F, Rostamzadeh A, Glanz W, Schulze JB, Schiebler SLF, Buerger K, Janowitz D, Perneczky R, Rauchmann BS, Teipel S, Kilimann I, Laske C, Munk MH, Spottke A, Roy-Kluth N, Wagner M, Frommann I, Lüsebrink F, Dechent P, Hetzer S, Scheffler K, Kleineidam L, Stark M, Schmid M, Ersözlü E, Brosseon F, Ewers M, Schott BH, Düzel E, Ziegler G, **Mattern H**, Schreiber S, Bernal J.
Inferior Frontal Sulcal Hyperintensities On Brain MRI Are Associated With Amyloid Positivity Beyond Age – Results From The Multicentre Observational DELCODE Study
Diagnostics 2024
DOI: 10.3390/diagnostics14090940
- J17 Morton L, Arndt P, Garza AP, Henneicke S, **Mattern H**, Gonzalez M, Dityatev A, Yilmazer-Hanke D, Schreiber S, Dunay IR.
Spatio-temporal dynamics of microglia phenotype in human and murine cSVD: impact of acute and chronic hypertensive states
Acta Neuropathologica Communications 2023
DOI: 10.1186/s40478-023-01672-0
- J18 Sangal M, Anikeeva M, Priese CS, **Mattern H**, Hövener J-B, Speck O.
MR based magnetic susceptibility measurements of 3D printing materials at 3 Tesla
Journal of Magnetic Resonance Open 2023
DOI:10.1016/j.jmro.2023.100138
- J19 Garcia-Garcia B*, **Mattern H***, Vockert N, Yakupov R, Schreiber F, Spallazzi M, Perosa V, Haghighia A, Speck O, Düzel E, Maass A, Schreiber S.
Vessel Distance Mapping: A novel methodology for assessing vascular-induced cognitive resilience
NeuroImage 2023
DOI:10.1016/j.neuroimage.2023.120094
***equal contribution**
- J20 Schreiber S, Bernal J, Arndt P, Schreiber F, Müller P, Morton L, Braun-Dullaeus RC, Valdés-Hernández MC, Duarte R, Wardlaw JM, Meuth SG, Mietzner G, Vielhaber S, Dunay IR, Dityatev A*, Jandke S*, **Mattern H***.
Brain Vascular Health in ALS Is Mediated through Motor Cortex Microvascular Integrity
Cells 2023
DOI:10.3390/cells12060957
***equal contribution**

- J21 Perosa V, Rotta J, Yakupov R, Kuijf HJ, Schreiber F, Oltmer JT, **Mattern H**, Heinze HJ, Düzel E, Schreiber S.
Implications of quantitative susceptibility mapping at 7 Tesla MRI for microbleeds detection in cerebral small vessel disease
Frontiers in Neurology 2023
DOI: 10.3389/fneur.2023.1112312
- J22 Spitz L, Gaidzik F, Stucht D, **Mattern H**, Preim B, Saalfeld S.
A hybrid hierarchical strategy for registration of 7T TOF-MRI to 7T PC-MRI intracranial vessel data
International Journal of Computer Assisted Radiology and Surgery 2023
DOI: 10.1007/s11548-023-02836-y
- J23 Chatterjee S, Prabhu L, Pattadkal M, Bortsova G, Sarasaen C, Dubost F, **Mattern H**, de Bruijne M, Speck O, Nürnberger A.
DS6: Deformation-Aware Semi-Supervised Learning: Application to Small Vessel Segmentation with Noisy Training Data
Journal of Imaging 2022
DOI: 10.3390/jimaging8100259
- J24 Bollmann S, **Mattern H**, Bernier M, Robinson SR, Park D, Speck O, Polimeni JR.
Imaging of the pial arterial vasculature of the human brain in vivo using high-resolution 7T time-of-flight angiography
eLife, 2022
DOI: 10.7554/eLife.71186
- J25 Perosa V, Arts T, Assmann A, **Mattern H**, Speck O, Oltmer J, Heinze H-J, Düzel E, Schreiber S, Zwanenburg JJM.
Pulsatility index in the basal ganglia arteries increases with age in elderly with and without cerebral small vessel disease
American Journal of Neuroradiology, 2022
DOI: 10.3174/ajnr.A7450
- J26 Iamshchinina P, Kaiser D, Yakupov R, Hänelt D, Sciarra A, **Mattern H**, Lüsebrink F, Düzel E, Speck O, Weiskopf N, Cichy R.
Perceived and mentally rotated contents are differentially represented in cortical layers of V1
Communications Biology, 2021
DOI: 10.1038/s42003-021-02582-4
- J27 Sciarra A, **Mattern H**, Yakupov R, Chatterjee S, Oeltze-Jafra S, Speck O.
Quantitative Evaluation of Prospective Motion Correction in Healthy Subjects at 7T MRI
Magnetic Resonance in Medicine, 2021
DOI: 10.1002/mrm.28998
- J28 Vockert N, Perosa V, Ziegler G, Schreiber F, Priester A, Spallazzi M, Garcia-Garcia B, Aruci M, **Mattern H**, Haghighi A, Düzel E, Schreiber S, Maass A.
Hippocampal vascularization patterns exert local and distant effects on brain structure but not vascular pathology in old age
Brain Communications, 2021
DOI: 10.1093/braincomms/fcab127
- J29 Lüsebrink F, **Mattern H**, Yakupov R, Acosta-Cabronero J, Ashtarayeh M, Oeltze-Jafra S, Speck O.
Comprehensive ultrahigh resolution whole brain in vivo MRI dataset as a human phantom
Scientific Data, 2021
DOI: 10.1038/s41597-021-00923-w

- J30 **Mattern H**, Knoll M, Lüsebrink F, Speck O.
Chemical sHift bAsed pRospective k-Space anonyMizAtion (CHARISMA)
 Magnetic Resonance in Medicine, 2020
 DOI: 10.1002/mrm.28460
- J31 Gretsche F*, **Mattern H***, Gallichan D, Speck O.
Fat navigators and Moiré phase tracking comparison for motion estimation and retrospective correction
 Magnetic Resonance in Medicine, 2019
 DOI: 10.1002/mrm.27908
***equal contribution**
- J32 Betts MJ, Kirilina E, Otaduy M, Ivanov D, Acosta-Cabronero J, Callaghan M, Lambert C, Cardenas-Blanco A, Pine K, Passamonti L, Loane C, Keuken MC, Trujillo P, Lüsebrink F, **Mattern H**, Liu K, Priovoulos N, Fließbach K, Dahl MJ, Maass A, Madelung CF, Meder D, Ehrenberg AJ, Speck O, Weiskopf N, Dolan R, Inglis B, Tosun D, Morawski M, Zucca ZA, Siebner HR, Mather M, Uludag K, Heinsen H, Poser BA, Howard R, Zecca L, Rowe JB, Grinberg LT, Jacobs HIL, Düzel E, Hämmerer D.
Locus coeruleus imaging as a biomarker for noradrenergic dysfunction in neurodegenerative diseases
 Brain, 2019
 DOI: 10.1093/brain/awz193
- J33 **Mattern H**, Sciarra A, Lüsebrink F, Acosta-Cabronero J, Speck O.
Prospective motion correction improves high resolution quantitative susceptibility mapping at 7T
 Magnetic Resonance in Medicine, 2018
 DOI: 10.1002/mrm.27509
- J34 Acosta-Cabronero J, Milovic C, **Mattern H**, Tejos C, Speck O, Callaghan MF.
A robust multi-scale approach to quantitative susceptibility mapping
 NeuroImage, 2018
 DOI: 10.1016/j.neuroimage.2018.07.065
- J35 Milovic C, Acosta-Cabronero J, Pinto JM, **Mattern H**, Andia M, Uribe S, Tejos C.
A new discrete dipole kernel for quantitative susceptibility mapping
 Magnetic resonance imaging, 2018
 DOI: 10.1016/j.mri.2018.04.004
- J36 **Mattern H**, Sciarra A, Godenschweger F, Stucht D, Lüsebrink F, Rose G, Speck O.
Prospective motion correction enables highest resolution time-of-flight angiography at 7T
 Magnetic Resonance in Medicine, 2017
 DOI: 10.1002/mrm.27033
- J37 Lüsebrink F, Sciarra A, **Mattern H**, Yakupov R, Speck O.
T1-weighted in vivo human whole brain MRI dataset with an ultrahigh isotropic resolution of 250 μm
 Scientific Data, 2017
 DOI: 10.1038/sdata.2017.32
- J38 Yarach U, In MH, Chatnuntawech I, Bilgic B, Godenschweger F, **Mattern H**, Sciarra A, Speck O.
A Model-based Iterative Reconstruction for Single-shot EPI at 7T
 Magnetic Resonance in Medicine, 2017
 DOI: 10.1002/mrm.26633

Letters & comments

- L1 Lenz C, Bauer M, Langkammer C, **Mattern H**, Santini F.
ESMRMB 2025 focus topic – cycle of translation
Magnetic Resonance Materials in Physics, Biology and Medicine, 2025
DOI:10.1007/s10334-025-01265-z
- L2 **Mattern H**
Editorial for: Subject discomfort of 5T MRI examinations and associated contributing factors
Journal of Magnetic Resonance Imaging, 2025
DOI:10.1002/jmri.29774
- L3 Müller P, Herzog M, Duderstadt Y, Kunz M, Lechner K, Meyer F, Schmeißer A, Meißler S, Ahrens D, Neumann K, **Mattern H**, Speck O, Behme D, Dunay I, Seeland U, Schreiber S, Braun-Dullaues R.
Kardiovaskuläre Prävention in Sachsen-Anhalt - Notwendigkeit und neue Perspektiven
Die Innere Medizin, 2024
DOI:10.1007/s00108-024-01789-x
- L4 Pinto J, McGee A, **Mattern H**, Markenroth Bloch K, Haast RAM, Küstner T, Vannesjo SJ.
ESMRMB 2024 focus topic “MR Beyond Structures: The dynamic body at different scales”
Magnetic Resonance Materials in Physics, Biology and Medicine, 2024
DOI:10.1007/s10334-024-01175-6
- L5 Schreiber S, Arndt P, Meuth SG, Dityatev A, **Mattern H**.
Brain microvascular disease and functional network connectivity – a call for a stage-based approach
Brain Communications, 2023
DOI:10.1093/braincomms/fcad135
- L6 Schreiber S*, John A-C*, Werner C, Vielhaber S, Heinze H-J, Speck O, Würfel J, Behme D*, **Mattern H***.
Counteraction of inflammatory activity in CAA-related subarachnoid hemorrhage
Journal of Neurology, 2022
DOI:10.1007/s00415-022-11437-9
***equal contribution**
- L7 Schreiber S, **Mattern H**.
Semi-automated Segmentation and Quantification of Perivascular Spaces at 7 Tesla in COVID-19
Qeios, 2022
DOI: 10.32388/Z0HEWQ

Workshops & Challenges

- X1 *SMILE-UHURA : Small Vessel Segmentation at Mesoscopic ScaLE from Ultra-High ResolUtion 7T Magnetic Resonance Angiograms*
Organizers: Chatterjee S, **Mattern H**, Dubost F, Schreiber S, Nürnberger A, Speck O.
Conference: IEEE - ISBI 2023, Cartagena de Indias, Colombia / hybrid
DOI: 10.7303/syn47164761 , Challenge homepage

Conference papers

- C1 Chatterjee S, Prabhu K, Pattadkal M, Bortsova G, Sarasaen C, Dubost F, **Mattern H**, de Bruijne M, Speck O, Nürnberger A
DS6, Deformation-aware Semi-supervised Learning: Application to Small Vessel Segmentation with Noisy Training Data
Medical Imaging with Deep Learning, 2021
Link to online version

- C2 Preeth CJ, **Mattern H**, Juneja M, Vogt J, Speck O, Hartkens T
Entropy based SVM Classifier for Automatic Detection of Motion Artifacts in Clinical MRI
 Bildverarbeitung für die Medizin, 2020
 DOI: 10.1007/978-3-658-29267-6_23
- C3 Thoma N, Odenbach R, **Mattern H**, Friebe M
Remotely controllable phantom rotation system for ultra-high field MRI to improve Cross Calibration
 Current Directions in Biomedical Engineering, 2019
 DOI: 10.1515/cdbme-2019-1570538325

Conference abstracts

- A1 **Mattern H**, Uher D, Tung Y-H, Sampaiores D, Maass A, Albrecht A, Hummel T, Speck O, Rothermel M; *Mapping the human anterior olfactory nucleus at 7 T using high-performance gradients: Preliminary results of the 7AON study*; 34th Annual Meeting of the International Society for Magnetic Resonance in Medicine. ISMRM, May 2026, Cape Town
- A2 Zhao M, Zou A, Hoinkiss DC, Bauer M, **Mattern H**, Assili S, Clement P, Pinto J, Ipek O, Hertanu A, Lena B, Haast R; *Echoes Beyond the Scanner: An Evolving Journey of the ESMRMB Podcast, Now Resonating in Cape Town*; 34th Annual Meeting of the International Society for Magnetic Resonance in Medicine. ISMRM, May 2026, Cape Town
- A3 Uher D, **Mattern H**, Tung Y-H, Yi Y-J, Düzel E, Hämmerer D, Speck O; *Comparison of complex denoising strategies via the stability and reproducibility of in-vivo NODDI metrics within hippocampus at 7T*; 34th Annual Meeting of the International Society for Magnetic Resonance in Medicine. ISMRM, May 2026, Cape Town
- A4 Akrasirakul S, **Mattern H**, Speck O, Yarach U; *Zero-Shot Self-Supervised with Channel Attention Mechanism for Motion-Resolved Radial Abdominal MRI Reconstruction*; 34th Annual Meeting of the International Society for Magnetic Resonance in Medicine. ISMRM, May 2026, Cape Town
- A5 Einspaenner E, Fuchs E, **Mattern H**, Grossmann H, Khadhraoui E, Muller SJ, Behme D; *Multi-modal assessment of acute sleep deprivation effects in healthy adults using 7T MRI*; Radiological Society of North America (RSNA) Annual Meeting 2025, Chicago
- A6 Mietzner G, Lümekemann L, Schreiber F, Brüggemann J, Benramadan A, Al-Dubai M, Sciarra A, Knoll C, Kuehn E, Speck O, Schreiber S, **Mattern H**; *Assessing arterial patterns in the primary motor cortex with 7 Tesla magnetic resonance imaging and vessel distance mapping*; 98th Congress of the Deutschen Gesellschaft für Neurologie. DGN, November 2025, Berlin
- A7 Engel J, Beck C, Bernal J, Fuchs E, Arndt P, Meuth SG, Behme D, Pawlitzki M, **Mattern H**, Schreiber S, Neumann K; *Perivaskuläre Räume der grauen Hirnsubstanz als möglicher Risikomarker einer ARIA unter Amyloid- β -Antikörpertherapie*; 98th Congress of the Deutschen Gesellschaft für Neurologie. DGN, November 2025, Berlin
- A8 Zhao M, Hoinkiss D, Bauer M, **Mattern H**, Assili S, Clement P, Pinto J; *From Magnetic Gradients to Sound Waves: A Five-Year Journey of the ESMRMB Podcast*; 41st Annual Scientific Meeting of European Society for Magnetic Resonance in Medicine and Biology. ESMRMB, October 2025, Marseille; DOI: s10334-025-01278-8
- A9 Einspanner E, **Mattern H**, Fuchs E, Müller S, Khadhraou E, Behme D; *Sleepy QSM: Study to assess the effect of sleep deprivation on brain homeostasis with 7T QSM and $qT1$* ; 41st Annual Scientific Meeting of European Society for Magnetic Resonance in Medicine and Biology. ESMRMB, October 2025, Marseille; DOI: s10334-025-01278-8

- A10 Marquardt J, Vockert N, Behrenbruch N, Schumann-Werner B, Hochkeppeler A, Büchel AT, Molloy EN, Schwarck S, Fischer L, Incesoy EI, Garcia-Garcia B, **Mattern H**, Morgado B, Esselmann H, Stephens AW, Schildan A, Barthel H, Sabri O, Wiltfang J, Kreissl MC, Düzel E, Kuehn E, Schreiber S, Maass A, Diersch N; *Hippocampal vascularization is associated with greater efficiency during a remote real world wayfinding training in older adults*; Alzheimer's Association International Conference (AAIC); Toronto, July 2025; DOI: 10.1002/alz70860_105004
- A11 Piechowiak C, Müller P, Bernal Moyano J, Kunz NA, Al-Zawity S, Hubert P, Nathania BP, Kunz M, Lading Y, Duarte R, Valdes Hernandez MdC, Wardlaw JM, **Mattern H**, Behme D, Neumann K, Braun-Dullaeus R, Schreiber S; *Acute physical exercise can exert measurable changes on perivascular spaces volumetry*; Alzheimer's Association International Conference (AAIC); Toronto, July 2025; DOI: 10.1002/alz70855_101885
- A12 Penalba Sanchez L, Yi YY, Kurt E, Femminella GD, Loane C, Duckett M, Callaghan M, Weiskopf N, Dolan R, Glanz W, Butryn M, **Mattern H**, Leiman M, Mann I, Pankratz U, Lübeck CY, Kelling N, Howard R, Düzel E, Hämmerer D; *Investigating the Role of the Locus Coeruleus Noradrenergic System in Cognitive Function in Amnesic Mild Cognitive Impairment: An fMRI Study*; European Psychiatry 68.S1 (2025): S836-S837; April 2025; DOI: 10.1192/j.eurpsy.2025.1697
- A13 Tung Y-H, **Mattern H**, Speck O; *Rapid Mesoscopic Resolution DTI with High-Performance 7T*; 33rd Annual Meeting of the International Society of Magnetic Resonance in Medicine. ISMRM, May 2025, Honolulu
- A14 Yarach U, Akrasirakul S, Godenschweger F, **Mattern H**, Tung Y-H, Speck O; *An Efficient Signal Averaging with Model-Based Reconstruction for Meso-scale Resolution T2*W GRE-EPI at 7T with High-Performance Gradients*; 33rd Annual Meeting of the International Society of Magnetic Resonance in Medicine. ISMRM, May 2025, Honolulu
- A15 Yarach U, Akrasirakul S, **Mattern H**, Speck O; *Improving T2-Weighted TGSE-BLADE Liver Imaging via Model-Based Reconstruction with Self-Breathing Motion Correction*; 33rd Annual Meeting of the International Society of Magnetic Resonance in Medicine. ISMRM, May 2025, Honolulu
- A16 Stirnberg R, **Mattern H**, Speck O, Stöcker T; *Ultra-fast whole-brain 3D-EPI across scales: pushing high-performance gradients at 7T*; 33rd Annual Meeting of the International Society of Magnetic Resonance in Medicine. ISMRM, May 2025, Honolulu
- A17 Huber R, Morgan T, Pizzuti A, Poser B, Dymerska B, Bosch D, Chai Y, Gudino N, Sarlls J, Dresbach S, Vu J, Evans J, Torrisi S, Stirnberg R, Tourell M, Bollmann S, Jin J, Vizioli L, Yacoub E, Priovoulos N, Brouwer E, van der Zwaag W, **Mattern H**, Tse D, Gomez D, Blazejewska A, Polimeni J, Bandettini P; *Should I do 7T fMRI with the pTx or sTx coil? The biggest difference is geometry*; 33rd Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, May 2025, Honolulu
- A18 Bollmann S, **Mattern H**, Gulban F, Speck O, Stöcker T, Stirnberg R; *Fast Mesoscopic Vascular Imaging Utilizing a High-Performance Gradient and a Segmented 3D EPI Readout*; ISMRM Joint Workshop of the Ultra-High Field MR & Brain Function Study Groups, March 2025, Annapolis;
- A19 Penalba Sanchez L, Yi Y-Y, Kurt E, Femminella GD, Loane C, Duckett M, Callaghan M, Weiskopf N, Dolan R, Glanz W, Butryn M, **Mattern H**, Leiman M, Mann I, Pankratz U, Lübeck CY, Kelling N, Howard R, Düzel E, Hämmerer D; *Investigating the Role of the Locus Coeruleus Noradrenergic System in Cognitive Function in Amnesic Mild Cognitive Impairment: An fMRI Study*; 33rd European Congress for Psychiatry. EPA, April 2025, Madrid;
- A20 **Mattern H**, Einspänner E, Fuchs E, Schreiber S, Behme D; *Sleepy QSM: Pilot study to assess the effect of sleep deprivation on brain homeostasis with 7T QSM*; 40th Annual Scientific Meeting of European Society for Magnetic Resonance in Medicine and Biology. ESMRMB, October 2024, Barcelona; DOI: 10.1007/s10334-024-01191-6

- A21 Akrasirakul S, Setakornnukul J, Puangragsa U, Sathitwatthanawirod C, **Mattern H**, Speck O, Yarach U; *Tikhonov Regularization with High SNR 3D Priors on Highly Accelerated 4D Body MRI* ; 40th Annual Scientific Meeting of European Society for Magnetic Resonance in Medicine and Biology. ESMRMB, October 2024, Barcelona; DOI: 10.1007/s10334-024-01191-6
- A22 Díaz M, Fuchs E, **Mattern H**, Behme D, Duarte R, Valdés-Hernández MC, Wardlaw J, Schreiber S, Trujillo M, Bernal J; *Contrast-agnostic deep-learning-based detection of perivascular spaces in magnetic resonance imaging*; Alzheimer's Association International Conference, July 2024, Philadelphia
- A23 Xu M, Ribeiro FL, Liu S, Shaw TB, **Mattern H**, Chatterjee S, Speck O, Gulban OF, Hartung G, Bollmann S, Polimeni JR, Barth M, Bollmann S; *VesselBoost: Combining Data Augmentation and Test-time Adaptation to Improve Segmentation of Small Vessels in Human Brain Magnetic Resonance Angiograms*; 2024 OHBM Annual Meeting, June 2024, Seoul; Aperture Neuro. 2024;4(Suppl 1). DOI: 10.52294/001c.120595
- A24 Vockert N, Lümke L, Yi YJ, Garcia-Garcia B, Behrenbruch N, Marquardt J, Hayek D, **Mattern H**, Hämmerer D, Diersch N, Kühn E, Schreiber S, Maass A; *Examining hippocampal vessel distances in relation to cognition and brain structure in young adults*; 2024 OHBM Annual Meeting, June 2024, Seoul; Aperture Neuro. 2024;4(Suppl 1). DOI: 10.52294/001c.120595
- A25 Müller P, Neumann K, Horndasch L, **Mattern H**, Groschek T, Meuth SG, Dunay I, Schmeißer A, Behme D, Schreiber S, Braun-Dullaeus R; *Cerebral small vessel disease as a mediator between heart failure with preserved ejection fraction and cognitive decline*; 10th European Stroke Organisation Conference, May 2024, Basel
- A26 Müller P, Horndasch L, Neumann K, **Mattern H**, Cardace S, Arndt P, Pfister M, Groschek T, Vielhaber S, Meuth SG, Dunay I, Schmeißer A, Behme D, Schreiber S, Braun-Dullaeus R; *Cerebral small vessel disease mediates the effect of arterial stiffness on cognitive decline in patients with heart failure with preserved ejection fraction*; 33rd European Meeting on Hypertension and Cardiovascular Protection, May 2024, Berlin; Journal of Hypertension. 42(Suppl 1):e92-e93; DOI: 10.1097/01.hjh.0001020204.96874.a6,
- A27 **Mattern H**, Tung YH, Lüsebrink F, Speck O; *Assessing the potential of 7T and high-performance gradients for high-resolution R2* mapping in deep gray matter*; 32nd Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, May 2024, Singapore
- A28 Tung YH, **Mattern H**, Speck O; *High b-value in vivo Whole-brain Diffusion MRI at 7T with a High-performance Gradient System*; 32nd Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, May 2024, Singapore
- A29 Velasquez Vides JR, Herrmann CJJ, Gladysz T, Shalika S, Millward JM, Waiczies S, Seeliger E, **Mattern H**, Rose G, Niendorf T; *Simultaneous and Respiratory Motion-Synchronized T2 and T2* Mapping of the Human Kidneys*; 32nd Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, May 2024, Singapore
- A30 Müller P, Horndasch L, **Mattern H**, Neumann K, Cardace S, Arndt P, Pfister M, Groschek T, Meuth SG, Dunay I, Schmeißer A, Behme D, Schreiber S, Braun-Dullaeus R; *Cerebral small vessel disease in patients with heart failure with preserved ejection fraction (HFpEF). A pilot study*; 90. Jahrestagung der Deutsche Gesellschaft für Kardiologie – Herz- und Kreislaufforschung e.V. (German Cardiac Society), April 2024, Mannheim DOI: 10.1007/s00392-024-02406-5
- A31 Stoll S, Lüsebrink F, Schwarzkopf DS, **Mattern H**, Liu P, Noelle J, Kühn E; *Mapping 2D Population Receptive Fields of the Fingertips in Human Primary Somatosensory Cortex*; 5th Brain In Depth Conference, March 2024, Tübingen
- A32 **Mattern H**, Schreiber S, Speck O; *Combining vessel distance mapping and non-negative matrix factorization to identify arterial and venous patterns in the putamen*; 2023 ICP Network Meeting Utrecht, November 2023, Utrecht

- A33 Mietzner G, Schreiber F, Lümekemann L, Brüggemann J, Sciarra A, Knoll C, Kuehn E, Speck O, Schreiber S, **Mattern H**; *Enabling in vivo assessment of motor cortex vessel dominance patterns using 7T MRI and vessel distance mapping*; 39th Annual Scientific Meeting of European Society for Magnetic Resonance in Medicine and Biology. ESMRMB, October 2023, Bern; DOI: 10.1007/s10334-023-01108-9
- A34 Fuchs E **Mattern H**, Vockert N, Arndt P, Neumann K, John A-C, Kühn E, Maass A, Düzel E, Schreiber S, Behme D; *Aging, cognition, and cerebral small vessel disease correlate with MR-based patterns of blood brain barrier breakdown*; 58. Jahrestagung der Deutschen Gesellschaft für Neuroradiologie e.V. Neurorad, October 2023, Kassel
- A35 Stoll S, Lüsebrink F, Schwarzkopf DS, **Mattern H**, Liu P, Kühn E; *Tailoring Population Receptive Field Modeling to Other Sensory Systems – From Vision to Touch*; System vision science summer school & symposium, August 2023, Tübingen
- A36 Behrenbruch N, Incesoy E, Bernal J, Menze I, Vockert N, Kleinedam L, Buerger K, Wolfsgruber S, Spottke A, Fleissbach K, Laske C, Pernecky R, Peters O, Priller J, Schneider A, Heneka MT, Wagner M, Teipel S, Wiltfang J, Speck O, Perosa V, Yakupov R, Jessen F, Düzel E, **Mattern H**, Schreiber S, Ziegler G, Maass A; *Altered Blood Oxygen Level Dependent Signal Fluctuations in the Spectrum of Alzheimer's Dementia and in Patients with White Matter Hyperintensities*; Alzheimer's Association International Conference, July 2023, Amsterdam
- A37 Stoll S, Lüsebrink F, Schwarzkopf DS, **Mattern H**, Liu P, Kühn E; *Modeling Population Receptive Fields of the Fingertips in Human Primary Somatosensory Cortex*; Touch 2023: Festival of Touch, July 2023, Marseille
- A38 **Mattern H**, Speck O; *The potential of ultra-high resolution 7T T2-weighted TSE to assess the glymphatic system's structure*; 31st Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, June 2023, Toronto
- A39 **Mattern H**, Lüsebrink F, Speck O; *The effect of imaging parameters, aging, and circadian rhythm on Freesurfer's estimates: A single subject study at 7T over 7 years*; 31st Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, June 2023, Toronto
- A40 Oltmer J, Beck J, **Mattern H**, Yakupov R, Auger C, Düzel E, van Veluw S, Schreiber S, Perosa V; *Enlarged Perivascular Spaces in the Basal Ganglia Surround Arteries, not Veins*; 31st Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, June 2023, Toronto
- A41 Chatterjee S, Gaidzik F, Sciarra A, **Mattern H**, Janiga G, Speck O, Nürnberger A, Pathiraja S; *Exploiting the inter-rater disagreement to improve probabilistic segmentation*; 31st Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, June 2023, Toronto
- A42 Chatterjee S, Chintalapati KV, Radhakrishna C, Kumar SCHR, Suttrave R, **Mattern H**, Speck O, Nürnberger A; *Enhancing Vessel Continuity in Deep Learning based Segmentation using Maximum Intensity Projection as Loss*; 31st Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, June 2023, Toronto
- A43 Besteher B, Kantola J, Li M, Bernal J, Machnik M, Reuken P, Finke K, Opel N, Kiviniemi V, **Mattern H**, Walter M; *From inflammation to degeneration – Enlarged perivascular spaces and glymphatic clearance in neuropsychiatric long-COVID syndrome*; Psychoimmunology Expert Meeting 2023, April 2023, Ulm; DOI: 10.1016/j.jadr.2023.100522
- A44 John A-C, Schreiber S, Werner C, Vielhaber S, Heinze H-J, Speck O, Würfel J, Behme D, **Mattern H**; *Evolution of cortical superficial siderosis in a 75-year-old patient with cerebral amyloid angiopathy*; 8th International Cerebral Amyloid Angiopathy (CAA) Conference, November 2022, Perth
- A45 Fuchs E, **Mattern H**, John A-C, Zubel S, Vielhaber S, Düzel E, Maass A, Kühn E, Schreiber S, Behme D; *Investigation of Blood Brain Barrier Breakdown and Early Clearance in Patients with Cerebral Small Vessel Disease using Contrast Enhanced MRI*; 57. Jahrestagung der Deutschen Gesellschaft für Neuroradiologie e.V. Neurorad, October 2022, Kassel

- A46 John A-C, Schreiber S, Werner C, Vielhaber S, Heinze H-J, Speck O, Würfel J, Behme D, **Mattern H**; *Entwicklung einer kortikalen superfiziellen Siderose bei einer 75-jährigen Patientin mit zerebraler Amyloidangiopathie (CAA)*; 57. Jahrestagung der Deutschen Gesellschaft für Neuroradiologie e.V. Neurorad, October 2022, Kassel
- A47 **Mattern H**, Speck O; *The potential of ultra-high resolution T2-weighted TSE acquired at 7T to assess structures of the glymphatic system*; ISMRM Workshop on Neurofluids. September 2022, Rome
- A48 Velasquez Vides JR, Speck O, Niendorf T, **Mattern H**; *Development of Freely Available 3D Radial Gradient Echo Sequences and Reconstruction*; 13th Annual Scientific Symposium Ultrahigh Field Magnetic Resonance. September 2022, Berlin
- A49 Garcia-Garcia B, **Mattern H**, Vockert N, Yakupov R, Schreiber F, Spallazzi M, Perosa V, Speck O, Düzel E, Maass A, Schreiber S; *Vessel distance mapping: a novel methodology for assessing vascular-induced cognitive resilience*; Alzheimer's Association International Conference, July 2022, San Diego; DOI: 10.1002/alz.063391
- A50 Ulbrich P, Morton L, Briese M, Lämmlein N, **Mattern H**, Hasanuzzaman M, Westhues M, Garz C, Becker A, Dityatev A, Jandke S, Yilmazer-Hanke D, Sendtner M, Dunay I, Schreiber S; *Stage-dependent responses of vascular and parenchymal cells in the hypertensive rat brain*; 31st European Meeting on Hypertension and Cardiovascular Protection, June 2022, Athens/Hybrid; DOI: 10.1097/01 .hjh.0000836812.14897.03
- A51 **Mattern H**, Angenstein F, Mawrin C, Perosa V; *Post mortem study of R2* and vessel distance maps across cortical depth*; Joint Annual Meeting ISMRM-ESMRMB 2022, May 2022, London, UK
- A52 **Mattern H**, Speck O; *Resolution-dependency of arterial and venous density estimates and vessel distance maps in deep gray matter*; Joint Annual Meeting ISMRM-ESMRMB 2022, May 2022, London, UK
- A53 Anikeeva M, Sangal M, Pravdivtseva MS, **Mattern H**, Speck O, Hövener J-B; *Magnetic Resonance Relaxometry and susceptibility of contemporary 3D printing materials*; First International Workshop on Reacting Particle-Gas Systems, June 2022, Bochum, Germany
- A54 Neumann K, Aruci M, Aki C, Günther M, Mistelbauer G, Garcia-Garcia B, Oeltze-Jafra S, Stucht D, **Mattern H**, Speck O, Perosa V, Düzel E, Schreiber S; *Vaskuläre Ursachen kognitiver Defizite bei Patienten mit zerebraler Mikroangiopathie*; Research Days 2021, Magdeburg, Germany
- A55 **Mattern H**; *Vessel distance mapping of the aging subcortical venous vasculature*; 37th Annual Scientific Meeting of European Society for Magnetic Resonance in Medicine and Biology. ESMRMB, October 2021, Online; DOI: 10.1007/s10334-021-00947-8
- A56 Bollmann S, **Mattern H**, Bernier M, Robinson SR, Park D, Speck O, Polimeni JR; *High resolution time-of-flight angiography of the pial arterial vasculature of the human brain in vivo*; 12th Annual Scientific Symposium Ultrahigh Field Magnetic Resonance. September 2021, Berlin, Germany
- A57 **Mattern H**; *Openly available sMall vEsSEL sEgmenTaTion pipelinE (OMELETTE)*; 29th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, May 2021, virtual meeting
- A58 **Mattern H**, Schreiber S, Speck O; *Vessel distance mapping for deep gray matter structures*; 29th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, May 2021, virtual meeting
- A59 Iamshchinina P, Kaiser D, Yakupov R, Haenelt D, Sciarra A, **Mattern H**, E. Düzel, Speck O, Weiskopf N, Cichy R; *Perceived and mentally rotated contents are differentially represented in cortical layers of V1*; 20th Annual Meeting Vision Sciences Society, October 2020; DOI: 10.1167/jov.20.11.766

- A60 **Mattern H**, Speck O; *Vessel distance mapping*; 36th Annual Scientific Meeting of European Society for Magnetic Resonance in Medicine and Biology. ESMRMB, September 2020, Online; DOI: 10.1007/s10334-020-00876-y
- A61 **Mattern H**, Knoll M, Lüsebrink F, Speck O; *Chemical sHift bAsed pRospectlve k-Space anony-MizAtion (CHARISMA)*; 28th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, August 2020, virtual meeting
- A62 **Mattern H**, Sciarra A, Dünnwald M, Chatterjee S, Müller U, Oeltze-Jafra S, Speck O; *Contrast prediction-based regularization for iterative reconstructions (PROSIT)*; 28th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, August 2020, virtual meeting
- A63 **Mattern H**, Odenbach R, Thoma N, Godenschweiger F, Speck O; *Remotely controllable phantom rotation device for cross-calibration at 7T*; 28th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, August 2020, virtual meeting
- A64 Lüsebrink F, **Mattern H**, Yakupov R, Oeltze-Jafra S, Speck O; *The human phantom: Comprehensive ultrahigh resolution whole brain in vivo single subject dataset*; 28th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, August 2020, virtual meeting data available at DOI: 10.24352/UB.OVGU-2020-145
- A65 Sciarra A, Dünnwald M, **Mattern H**, Speck O, Oeltze-Jafra S; *Super-Resolution with Conditional-GAN for MR Brain Images*; 28th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, August 2020, virtual meeting
- A66 Iamshchinina P, Kaiser D, Yakupov R, Haenelt D, Sciarra A, **Mattern H**, E. Duezel, Speck O, Weiskopf N, Cichy R; *Perceived and mentally rotated contents are differentially represented in cortical layers of V1*; 26th Annual Meeting of the Organization for Human Brain Mapping. OHBM, June 2020, virtual meeting
- A67 Lüsebrink F, **Mattern H**, Oeltze-Jafra S, Speck O; *Beyond high resolution: Denoising during image reconstruction to improve image quality*; 36th Annual Scientific Meeting of European Society for Magnetic Resonance in Medicine and Biology. ESMRMB, October 2019, Rotterdam; DOI: 10.1007/s10334-019-00755-1
- A68 Lüsebrink F, **Mattern H**, Oeltze-Jafra S, Speck O; *Image reconstruction pipeline*; 36th Annual Scientific Meeting of European Society for Magnetic Resonance in Medicine and Biology. ESMRMB, October 2019, Rotterdam; DOI: 10.1007/s10334-019-00756-0
- A69 Thoma N, Odenbach R, **Mattern H**, Friebe M; *Remotely controllable phantom rotation system for ultra-high field MRI to improve Cross-Calibration*; 53rd Annual Conference of the German Society for Biomedical Engineering, September 2019, Frankfurt am Main, Germany
- A70 Lüsebrink F, **Mattern H**, Oeltze-Jafra S, Speck O; *Denoising during image reconstruction to improve image quality of high resolution MR data*; 3rd Brain in Depth Symposium, April 2019, Leipzig, Germany
- A71 **Mattern H**, Acosta-Cabronero J, Speck O; *High resolution imaging of the arterial and venous vasculature in deep gray matter*; 27th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, May 2019, Montreal, Canada
- A72 Gretsche F, **Mattern H**, Gallichan D, Speck O; *Direct comparison of fat navigators and Moiré phase tracking for retrospective brain motion correction at 7T*; 27th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, May 2019, Montreal, Canada
- A73 Cardenas-Blanco A, Chen Y, Valdes-Herrera JP, Yakupov R, **Mattern H**, Sciarra A, Berron D, Maass A, Speck O, Düzel E; *Hippocampal subfield segmentation and partial volume effects - reliability assessment*; 27th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, May 2019, Montreal, Canada
- A74 **Mattern H**, Speck O; *Optimizing Cartesian compressed sensing for ultra-high resolution Time of Flight angiography*; Joint Annual Meeting ISMRM-ESMRMB 2018, June 2018, Paris, France

- A75 Sciarra A, **Mattern H**, Speck O; *Machine learning algorithms for detection of motion artifacts: a general approach*; Joint Annual Meeting ISMRM-ESMRMB 2018, June 2018, Paris, France
- A76 **Mattern H**, Odenbach R, Parsanejad P, Friebe M; *3D-printed MRI marker for personalized interventional applications through T1 and T2 relaxation time matching*; Computer Assisted Radiology and Surgery Congress and Exhibition. CARS, June 2018, Berlin, Germany; DOI: /10.1007/s11548-018-1766-y
- A77 Lüsebrink F, **Mattern H**, Speck O; *Beyond high resolution: A glimpse into the future*; 2nd Brain in Depth Symposium, May 2018, Magdeburg, Germany
- A78 Lüsebrink F, **Mattern H**, Sciarra A, Yakupov R, Speck O; *Ultrahoch aufgelöster T1-gewichteter in vivo MRT Datensatz des gesamten menschlichen Gehirns*; 20th Annual Meeting German Chapter of International Society of Magnetic Resonance in Medicine, November 2017, Göttingen; ISSN: 1863-6365
- A79 **Mattern H**, Sciarra A, Speck O; *Non-iterative, retrospective background suppression in time of flight angiography*; 34th Annual Scientific Meeting of European Society for Magnetic Resonance in Medicine and Biology. ESMRMB, October 2017, Barcelona; DOI: 10.1007/s10334-017-0632-1
- A80 **Mattern H**, Sciarra A, Speck O; *Wavelet entropy: quantifying small-scale head motion artifacts*; 34th Annual Scientific Meeting of European Society for Magnetic Resonance in Medicine and Biology. ESMRMB, October 2017, Barcelona; DOI: 10.1007/s10334-017-0632-1
- A81 **Mattern H**, Lüsebrink F, Speck O; *Ultrahochauflöste MRT des Gehirns mittels prospektiver Bewegungskorrektur*; 52. Jahrestagung der Deutschen Gesellschaft für Neuroradiologie e.V. Neurorad, October 2017, Cologne
- A82 **Mattern H**, Lüsebrink F, Sciarra A, Speck O; *More than meets the eye: Quantitative evaluation of prospective motion correction at 7T*; 25th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, April 2017, Honolulu, USA
- A83 **Mattern H**, Sciarra A, Godenschweger F, Stucht D, Lüsebrink F, Speck O; *Beyond the biological resolution limit: Prospectively motion corrected Time of Flight angiography at 7T*; 25th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, April 2017, Honolulu
- A84 Sciarra A, **Mattern H**, Yakupov R, Stucht D, Schulze P, Godenschweger F, Speck O; *Quantitative Evaluation of Prospective Motion Correction for Structural Imaging at 7T*; 25th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, April 2017, Honolulu
- A85 Cardenas-Blanco A, Berron D, Chen Y, **Mattern H**, Yakupov R, Sciarra A, Speck O, Düzel E; *Impact of Prospective Motion Correction in 7T fMRI Studies*; 25th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, April 2017, Honolulu
- A86 Rua C, In MH, Yakupov R, **Mattern H**, Costagli M, Symms M, Del Guerra A, Tosetti M, Speck O; *Study of the PSF Distortion Correction for Ultra-High Field BOLD fMRI*; 25th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, April 2017, Honolulu
- A87 Lüsebrink F, **Mattern H**, Sciarra A, Speck O; *Quantitative and Qualitative Evaluation of Bias Field Correction Methods*; 25th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, April 2017, Honolulu
- A88 Yarach U, **Mattern H**, Speck O; *Iterative SENSE with Integrated EPI Nyquist Ghost and Distortion Corrections*; 25th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, April 2017, Honolulu
- A89 **Mattern H**, Acosta-Cabronero J, Sciarra A, Speck O; *Prospectively motion-corrected QSM at 7 Tesla*; 33rd Annual Scientific Meeting of European Society for Magnetic Resonance in Medicine and Biology. ESMRMB, September 2016, Vienna; DOI: 10.1007/s10334-016-0570-3

- A90 **Mattern H**, Sciarra A, Godenschweger F, Stucht D, Lüsebrink F, Speck O; *Beyond high resolution: Prospectively motion corrected Time of Flight angiography with 150 μm isotropic resolution at 7T under SAR constraints*; 28th Annual International Conference of Society of Magnetic Resonance Angiography. SMRA, September 2016, Chicago
- A91 **Mattern H**, Acosta-Cabronero J, Sciarra A, Speck O; *Prospective motion correction for high-resolution QSM*; 4th International Workshop on Quantitative Susceptibility Mapping. September 2016, Graz
- A92 Sciarra A, **Mattern H**, Stucht D, Yakupov R, Schulze P, Godenschweger F, Speck O; *Prospective Motion Correction Applications at 7T*; ITN Network HiMR Final Meeting. June 2016, Magdeburg
- A93 Rua C, Costagli M, Biagi L, Symms M, Cosottini M, Wastling S, Barker G, In MH, Yakupov R, **Mattern H**, Speck O, Del Guerra A, Tosetti M; *Strategies for high-resolution fMRI at 7 Tesla*; ITN Network HiMR Final Meeting. June 2016, Magdeburg
- A94 **Mattern H**, Acosta-Cabronero J, Sciarra A, Speck O; *Prospectively motion corrected QSM-based venograms at 7 Tesla*; 7th Annual Scientific Symposium Ultrahigh Field Magnetic Resonance. June, 2016 Berlin
- A95 **Mattern H**, Sciarra A, Godenschweger F, Stucht D, Lüsebrink F, Speck O; *Prospective motion correction for ultra-high resolution Time of Flight angiography at 7T under SAR constraints*; 24th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, May 2016, Singapore
- A96 Lüsebrink F, Sciarra A, **Mattern H**, Yakupov R, Speck O; *Beyond High Resolution MPRAGE: In Vivo T1-Weighted Imaging at 7T with 250 μm Isotropic Resolution Using Prospective Motion Correction*; 24th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, May 2016, Singapore
- A97 Yarach U, **Mattern H**, Sciarra A, Speck O; *Combination of Individual Coil QSM at High Field Strength (7T)*; 24th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, May 2016, Singapore
- A98 Yarach U, Stucht D, **Mattern H**, Godenschweger F, Speck O; *Gradient Nonlinearity and B0-Induced Distortion Corrections of Prospective Motion Correction Data at 7T MRI*; 24th Annual Meeting of International Society of Magnetic Resonance in Medicine. ISMRM, May 2016, Singapore
- A99 **Mattern H**, Godenschweger F, Sciarra A, Stucht D, Speck O; *Ultra-high resolution Time of Flight angiography at 7T with prospective motion correction under SAR constraints*; 18th Annual Meeting German Chapter of International Society of Magnetic Resonance in Medicine. DS-ISMIR, November 2015, Münster; ISSN: 1863-6365

Supervision of students

- ongoing **Jascha Brüggemann**, *Vessel distance mapping to assess vascular resistance and resilience*, PhD thesis, supervisor.
- 2025 **Grazia Mietzner**, *Hochauflösende 7T-MRT zur nichtinvasiven Untersuchung der vaskulären Architektur des primären motorischen Kortex*, Medical doctorate thesis (Dr. med.), co-supervisor.
- 2025 **Simon Carl Priese**, *Vibrationsverhalten hochleistungsfähiger Gradientensysteme in der Ultrahochfeld-MRT mit Auswirkungen auf die Bildqualität*, Master thesis, examiner & supervisor.
- 2024 **Maryam Kargaran**, *Automatic vessel wall enhancement detection from contrast enhanced MRI*, Research project, examiner & supervisor.
- 2022 **Elnaz Khosroshahi**, *Investigation of vascular contributions to Alzheimer's disease with vessel distance mapping technique*, Master thesis, examiner & supervisor.
- 2020 **Jose Raul Velasquez Vides**, *Development of an freely available 3D radial gradient echo sequence and reconstruction*, Master thesis, examiner & supervisor.

- 2020 **Mahantesh Vishvanath Pattadkal & Kartik Prabhu**, *DS6: Deformation-aware learning for small vessel segmentation with small, imperfectly labeled dataset*, Team project, supervisor.
- 2019 **Ursula Müller**, *Compressed Sensing for high resolution Time of Flight angiographic Ultra High Field Magnetic Resonance Imaging*, Master thesis, examiner & supervisor.
- 2019 **Martin Quitmann**, *Variable k-space averaging of Inversion Recovery Sequences for high-resolution MR images with increased SNR*, Master thesis, examiner & supervisor.
- 2019 **Chandrakanth Jayachandran Preetha**, *Simulation of motion artifacts in magnetic resonance imaging*, Research project, examiner & supervisor.
- 2018 **Ursula Müller**, *Analysis of undersampling patterns for compressed sensing in magnetic resonance imaging*, Research project, examiner & supervisor.

Teaching

- since 2025 **Physics I & II**, *Course leader & instructor*, summer & winter term.
- since 2023 **Magnetic Resonance System Engineering**, *Tutor*, course provided to Siemens employees exclusively.
- 2019-2025 **Basics of Magnetic Resonance**, *Tutor*, winter term.
- 2018-2025 **Physics for Engineers**, *Tutor*, winter & summer term.

Invited talks

- IT1 *Magnetic Resonance Imaging in a Nutshell*, MODOLFOR Workshop, Magdeburg 06.10.2025
- IT2 *7T neurofluid imaging*, SynAGE Workshop, Magdeburg 24.10.2024
- IT3 *7T imaging of the cerebral structure & vasculature at the mesoscopic scale*, Inselspital-Universitätsspital Bern, 19.06.2024
- IT4 *7T MRI and Vessel Distance Mapping: Unveiling Mesoscopic Vascular Patterns*, Max Planck Institute for Human Development Berlin, 30.04.2024
- IT5 *High-resolution structural brain imaging*, lecture as part of the motion correction online series of British chapter of the ISMRM, 14.03.2024
- IT6 *Ultra-high field MRI*, lecture at the 54th annual meeting of the Deutschen Gesellschaft für Medizinische Physik (DGMP), 28.09.2023
- IT7 *Ultra-high field imaging to study the brain's vasculature and structure*, seminar of the Berlin Ultra High Field Facility, 16.11.2022
- IT8 *Ultra-high resolution structural and vascular MRI*, lecture for British chapter of the ISMRM online series, 14.10.2022
- IT10 *In-vivo imaging of vascular and (g)lymphatic system of the human brain with MRI*, SynAGE Workshop: Assessment of vascular and (g)lymphatic functions, Magdeburg, 05.05.2022
- IT11 *(f)MRI oversimplified*, lecture for the developmental psychology course at the University of Innsbruck, 16.03.2022
- IT15 *Robustness and Motion Correction: Hardware and Sequence Solutions*, 36th ESMRMB, Rotterdam, 05.10.2019
- IT16 *Bewegungskorrektur*, DGMP: 54. Arbeitskreistreffen, Erlangen, 23.09.2019
- IT17 *Motion corrected, high resolution MRI of the human anatomy and vasculature*, Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, 18.07.2019

Committee & Memberships

- 2025 Nomination and Awards Committee of the ESMRMB
- since 2023 Center for Behavioral Brain Sciences

2022-2025 Congress Program Committee for the 39th, 40th & 41st Annual Meeting of the ESMRMB
since 2021 ESMRMB Early Career Researchers ad hoc Committee
 ○ 2021-2023 Member
 ○ 2024 Vice-Chair
 ○ 2025 Chair

2021-2022 ISMRM Motion Correction Study Group Committee (trainee representative)

Language skills

German mother tongue

English fluent in speech and writing

Programming skills

since 2012 MATLAB

since 2019 Python

since 2015 C/C++

since 2015 IDEA